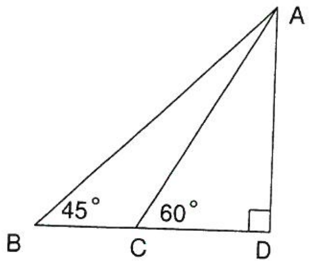
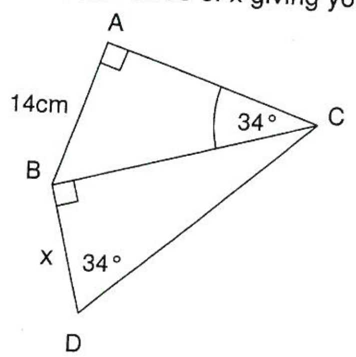
YEAR 11 TRIGONOMETRY REVISION

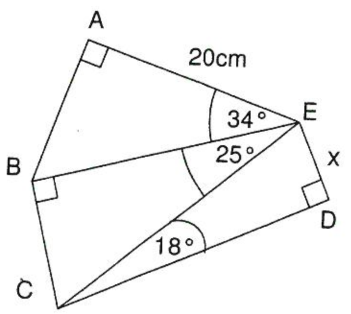
1. In the diagram below AC is 20 cm. Calculate the lengths (a) AD, (b) CD, (c) BC and (d) AB. Give non-integral answers correct to one decimal place.



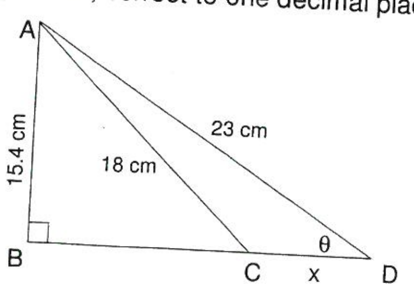
2. Find the value of x giving your answer correct to one decimal place.



3. Find the value of x giving your answer correct to one decimal place.



4. Find, correct to one decimal place, the angle marked θ and the length x.



5. A road is inclined so that it rises 1 metre for each horizontal distance of 10 metres. Calculate the angle the road makes with the horizontal.

6. In triangle ABC, a = 58 cm, b = 49 cm and $\angle C = 33°$.
(a) Find the length of the perpendicular from A to BC. Give answer correct to one decimal place.

(b) Hence find the size of angle A of triangle ABC. Give answer correct to the nearest tenth of a degree.

7. Nicole notes that the angle of elevation of the top of a 40 metre tower is 30°. She walks closer to the tower, stops and notes that the angle of elevation of the top of the tower is now 45°. How far did she walk?

8. From the top of a building an observer notes that the angles of depression of two objects in a line on the level ground are 45° and 25°. If the building is 60 metres high, how far apart are the two objects?

9. Two cars leave the same service station (S) at the same time. The first car (A) travels at an average speed of 60km per hour on a bearing of 062°T and the second car (B) travels at an average speed of 70km per hour on a bearing of 152°T.

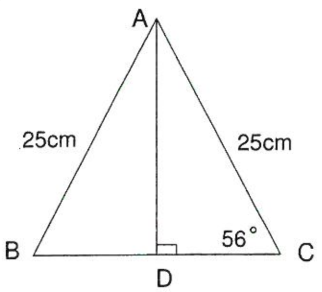
(a) Draw a diagram showing the service station and the location of both cars after one hour.

(b) Find the distance between the two cars after one hour. Give the answer to the nearest tenth of a kilometre.

(c) Find the bearing of the second car(B) from the first car(A) after one hour.

10. A hot air balloon rises vertically between two points P and Q. Observers at points P and Q note that at 10:00 am that the angles of elevation of the balloon are 65° and 35° respectively. The altitude of the balloon at 10:00 am is 300 metres. Calculate the distance between points P and Q.

11. Find the length of side BC in triangle ABC. Give your answer correct to two decimal places.



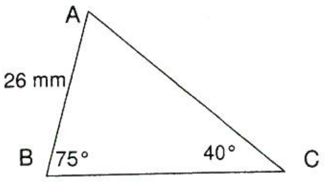
12. A kite is flying on a cord of length 210 metres. If the kite is 120 metres above the ground, what angle does the cord make with the ground?

13. In triangle ABC, $\angle A = 20°$, $\angle B = 120°$ and the length of AB is 6.8 m.

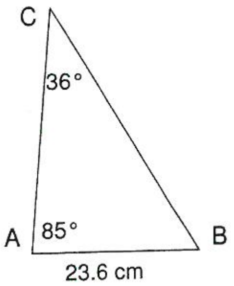
(a) Find the length of AC. (b) Find the length of BC.

14. The base of an isosceles triangle is 20 cm long and the base angles are each $70°$. Find the lengths of of the equal sides of this triangle using the sine rule. Give answer rounded to two decimal places.

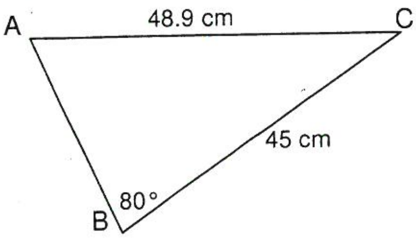
15. Solve ΔABC by finding the size of angle A and lengths of the sides AC and BC. Give non-integral answers correct to one decimal place.



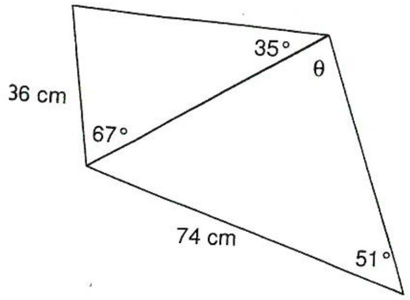
16. Solve ΔABC by finding the size of angle B and lengths of the sides AC and BC. Give non-integral answers correct to one decimal place.



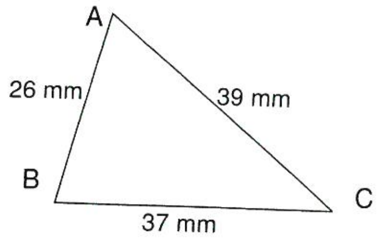
17. Solve acute ΔABC by finding the sizes of angles A and C and the length of the side AB. Give non-integral answers correct to one decimal place.



18. Find the size of the angle marked θ giving your answer correct to the nearest tenth of a degree.

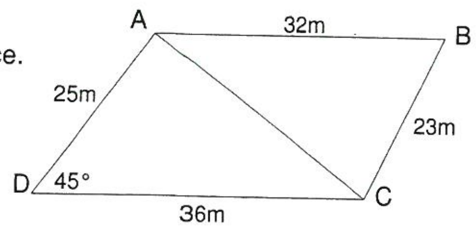


19. Solve ΔABC by finding the sizes of angles A, B and C. Give answers correct to one decimal place.



20. Consider quadrilateral ABCD shown right.

(a) Find the length of the diagonal AC correct to one decimal place.

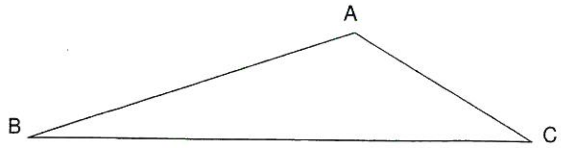


(b) Find the size of $\angle ABC$ to the nearest degree.

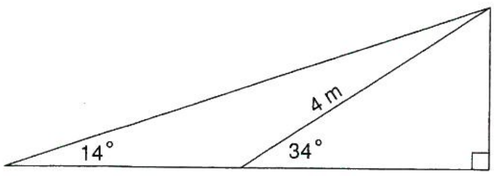
(c) Find the area of ΔADC correct to one decimal place.

(d) Calculate the area of quadrilateral ABCD to the nearest square metre.

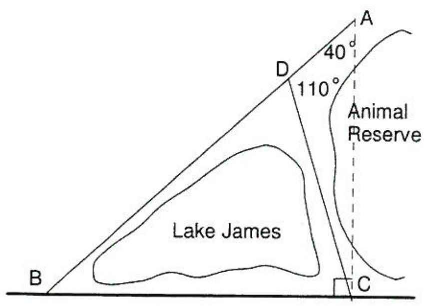
21. An architect is designing a roof for a garage, the cross section of the roof is shown opposite.
BC is 18 metres and $\angle ACB = 30°$.
If the architect wants angle BAC to be $130°$, what to the nearest centimetre must be the length of AC?



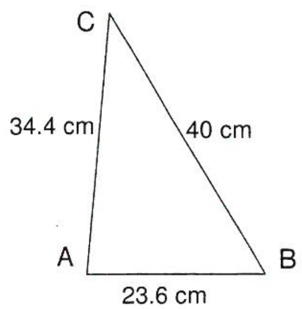
22. A 4 metre ramp has an angle of inclination of $34°$.
This ramp is found to be too steep. Find the length of a ramp with an angle of inclination of $14°$.
Give the ramp length to the nearest centimetre.



23. A 20 km road is to be built from A to C. C lies on an existing road BC.
Surveyors find that the proposed road will pass through an Animal Reserve and have to find an alternative route.
One option is to build road ADC and the other option is to build road ADB.
Find the length of road that needs to be built in each case and state which is the better option.

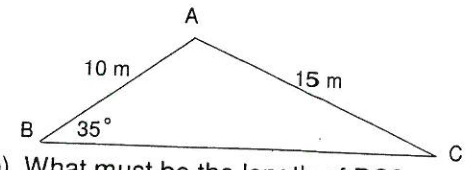


24. Solve ΔABC by finding the sizes of angles A, B and C. Give answers correct to one decimal place.



25. Oxford Street runs North-South and Vincent Street runs West-East. A motorist travelling west along Vincent Street and at a distance of 20 metres from the intersection of Oxford and Vincent Streets observes a speed camera situated in Oxford Street on a bearing of 312°T How far, to the nearest metre, is the speed camera from the intersection?

26. An architect is designing a roof for a garage, the cross section of the roof is shown opposite. AB is 10 metres and $\angle ABC = 35°$.

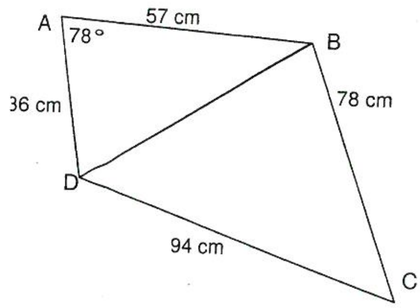


The architect wants AC to be 15 m.

(a) What must be the size of $\angle BAC$? Answer to nearest tenth of a degree.

(b) What must be the length of BC? Answer to nearest centimetre.

27. Consider quadrilateral ABCD shown right.



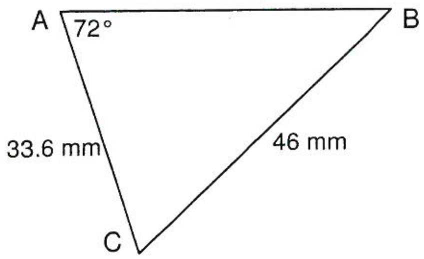
(a) Find the length of the diagonal BD correct to one decimal place.

(b) Find the size of $\angle BCD$ to the nearest degree.

(c) Find the area of ΔABD correct to one decimal place.

(d) Calculate the area of quadrilateral ABCD to the nearest square centimetre.

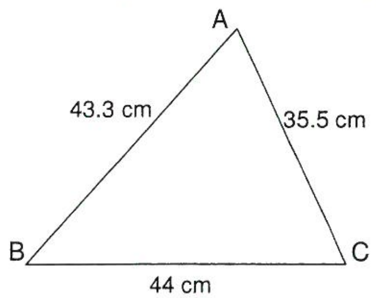
28. Solve acute ΔABC by finding the sizes of angles B and C and the length of the side AB. Give non-integral answers correct to one decimal place.



29. In ΔABC $a = 10$ cm, $b = 13$ cm and angle $B = 60°$. Find the remaining angles and side.

30. In ΔPQR, $\angle P = 23°$, $\angle Q = 75°$ and $PR = 200$ cm. Solve ΔPQR by finding the missing sides and angles.

31. Solve ΔABC by finding the sizes of angles A, B and C. Round answers to nearest tenth of a degree.



32. Solve ΔABC $a = 11$ cm, $b = 8$ cm and $c = 9$ cm. Give the solution correct to one decimal place.